DATA SCIENCE PROJECT

Exoplanet Habitability

Can we estimate how likely a known exoplanet is to be habitable — using only its orbital and host-star data?

NASA Exoplanet Archive · 5,578 confirmed planets · Python / scikit-learn

METHOD

How We Define "Habitable" Here

- **Habitable Zone** — distance from the star where liquid water could exist, based on the star's brightness
- **Rocky planet** — radius under 1.6 Earth radii (bigger = likely gas/ice giant)
- **Earth Similarity Index (ESI)** — a 0–1 score comparing radius, density, gravity, and temperature to Earth

A planet is flagged *potentially habitable* only if it meets **all three** conditions.

> This is a teaching heuristic, not a scientific verdict — atmosphere, magnetic field, and moons aren't measurable for most exoplanets yet.

DATA

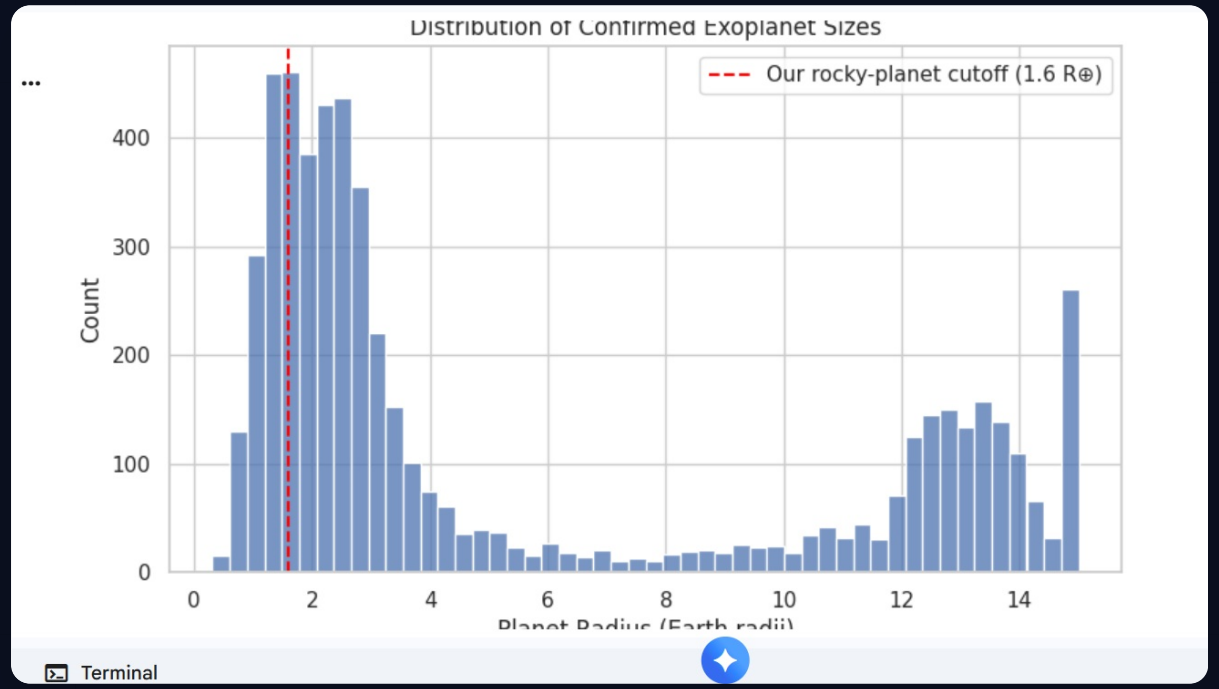
From 6,372 Planets to a Clean Dataset

Live data pulled from the NASA Exoplanet Archive. Rows missing essential values (distance, radius, star temperature) were dropped.

6,372 planets fetched	**5,578** after cleaning	**122** inside the Habitable Zone	**12** meet all 3 criteria

EXPLORATORY ANALYSIS

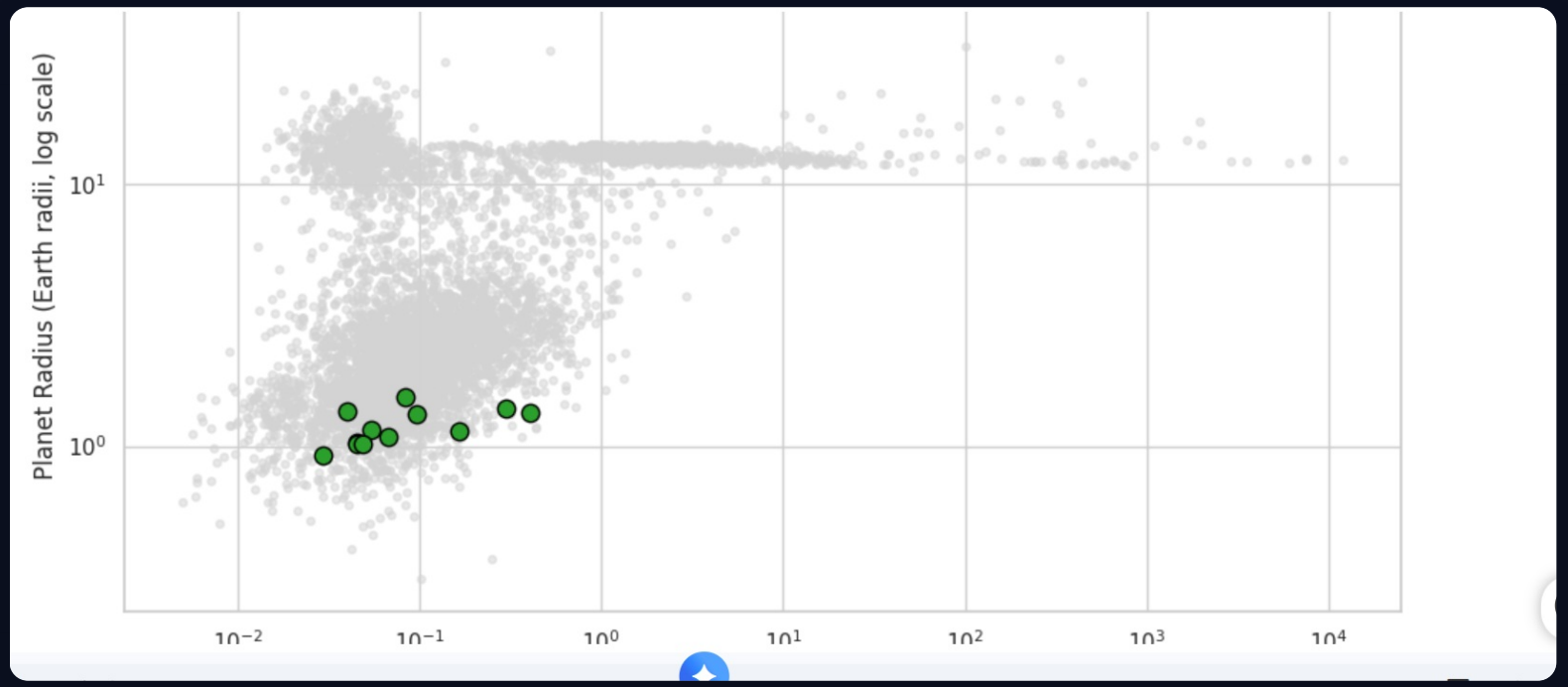
Planet Sizes Cluster in Two Groups



The gap around 4–8 Earth radii is a known phenomenon — the "Radius Valley" — where planets tend to end up either small & rocky, or large & gaseous.

RESULT

12 Candidates, Found Among 5,578 Planets



Green = our 12 candidates. Nearly all match well-known picks from the literature — Proxima Cen b, TRAPPIST-1e, TOI-700d, Kepler-442b.

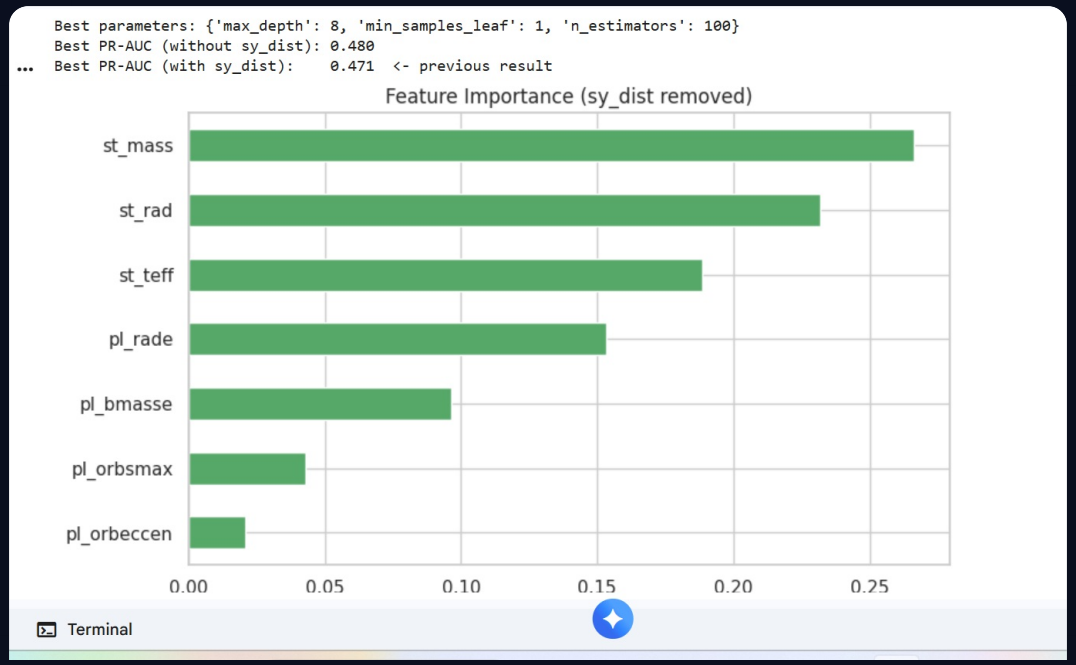
Can a Model Learn the Pattern?

A Random Forest was trained on raw stellar/orbital data only — not the engineered HZ/ESI features, to avoid leaking the answer.

0.48
PR-AUC (tuned model)

0.002
random-guess baseline

Star mass, radius, and temperature are the strongest predictors — consistent with the physics of the Habitable Zone.



CATCH #1 & #2

Two Surprises That Turned Out to Be Physics

> **The Radius Valley**
>
> Our 1.6 Earth-radius cutoff for "rocky" wasn't arbitrary — it landed right on a real, independently known gap in planet sizes (the "Radius Valley"), where planets tend to end up either small & rocky or large & gaseous, rarely in between. A nice confirmation that our threshold matches real physics.

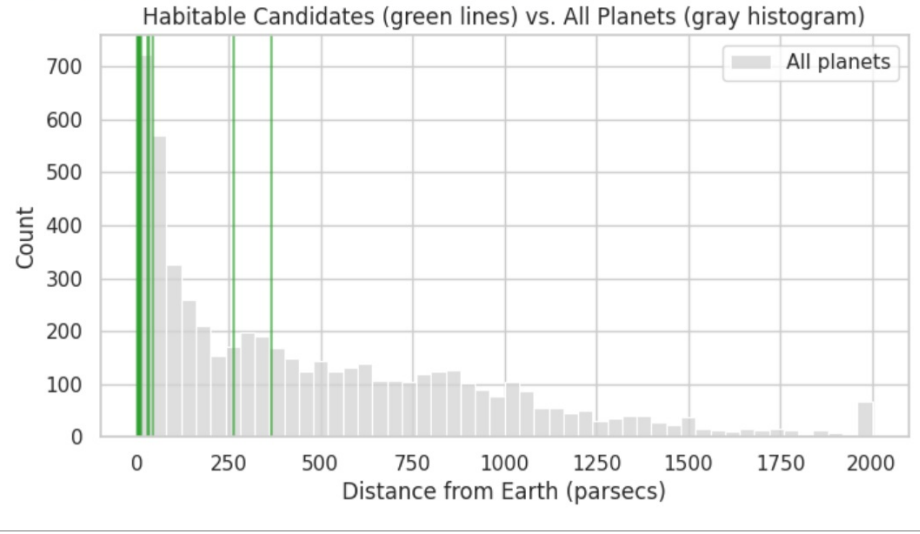
> **A Temperature Mismatch, Fixed**
>
> Some planets scored a near-perfect ESI yet sat outside the Habitable Zone — e.g. Kepler-438 b. The cause: we compared planet temperatures to Earth's warm surface temperature (288K), but our planet values were "bare" blackbody temperatures (no greenhouse effect). Earth's own blackbody temperature is 255K. Fixing the reference value realigned ESI with the Habitable Zone for most planets.

CATCH #3

Distance From Earth Was a Red Herring

```
plt.show()

Distance from Earth (sy_dist, parsecs):
  Habitable candidates: {'mean': 65.1, 'min': 1.3, 'max': 366.0}
  All planets:          {'mean': 495.2, 'min': 1.3, 'max': 4330.0}
```



Our candidates sit ~8x closer to Earth than the average planet — not physics, but detection bias: nearby planets are easier to measure precisely. Removing this feature improved the model.

CONCLUSIONS

What We Learned

- The heuristic correctly surfaces well-known habitable candidates — a solid sanity check
- A model can learn the pattern from raw data, led by real physics (star mass/radius/temperature)
- Two real pitfalls were caught and fixed: an inconsistent temperature reference, and a distance-based bias
- **Limitation:** only 12 positive examples — any model evaluation here carries real statistical uncertainty
- **Next steps:** atmosphere data, stellar flare activity, larger confirmed samples over time